

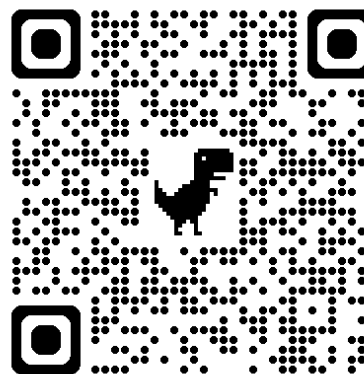


自然語言處理與應用

Natural Language Processing and Applications

LLM Introduction and RAG

Instructor: 林英嘉 (Ying-Jia Lin)
2026/05/04



GitHub



Slido
NLP0504
([Link](#))

Outline

- LLM Introduction [35min]
- Announcements [10min]
- Assignment 4 [50min]
- Retrieval-Augmented Generation (RAG) [35min]
- Quiz [15min]

LLM Introduction

Transformer 架構的後續應用

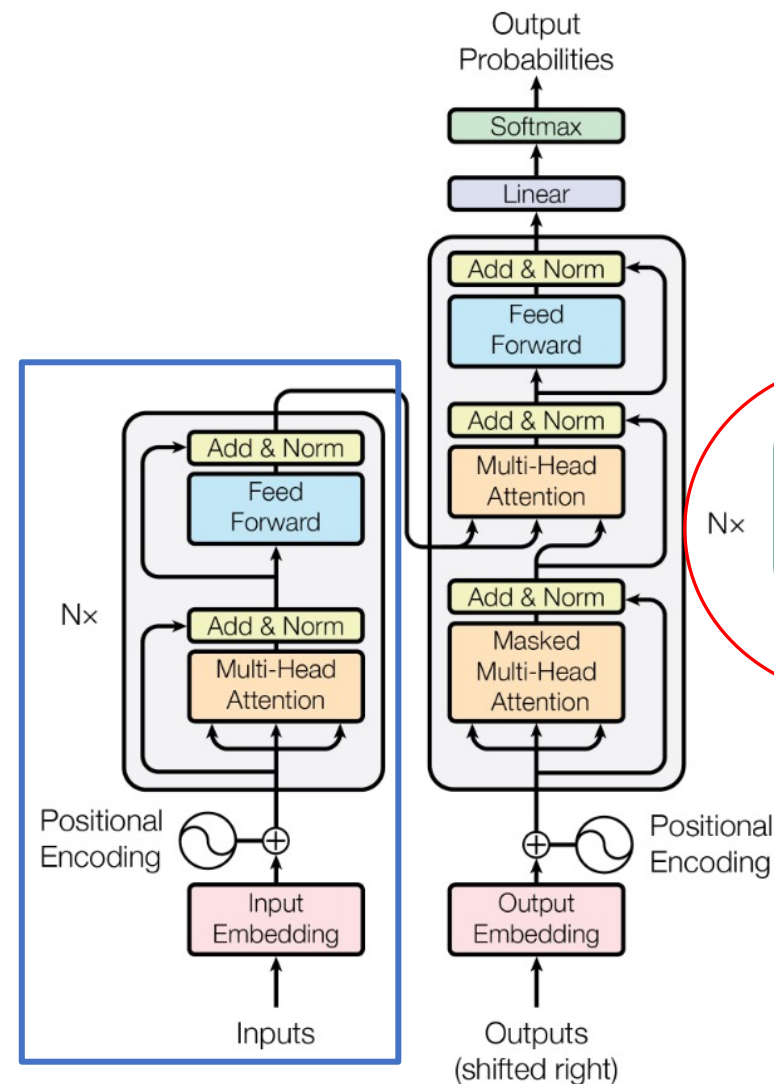
- Generative Pre-training (GPT) series 是 **Transformer Decoder**
- BERT (Bidirectional encoder representations from transformers) 是 **Transformer Encoder**
 - [Devlin et al., 2018](#)



(BERT)

N=12

Encoder



Nx

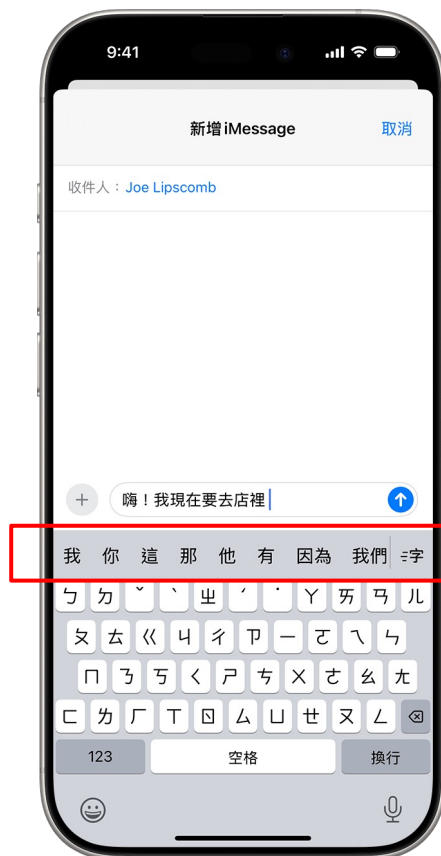
(GPT)

[Recap] GPT 是一個語言模型

<https://developers.google.com/machine-learning/resources/intro-llms>

- 語言模型定義：語言模型 (Language Model) 是一個能預測並生成語言的機器學習模型

(例子) 下一個字預測 ➡



先 Pre-training，再 Fine-tuning

Pre-training



Fine-tuning

在大量資料上進行訓練，通常是自監督式 (Self-Supervised Training)

下一個字預測

文字還原

...

在目標資料上 (Down-stream tasks) 進行監督式學習 (Supervised Learning)，資料規模通常較 Pre-training 階段小。

機器翻譯

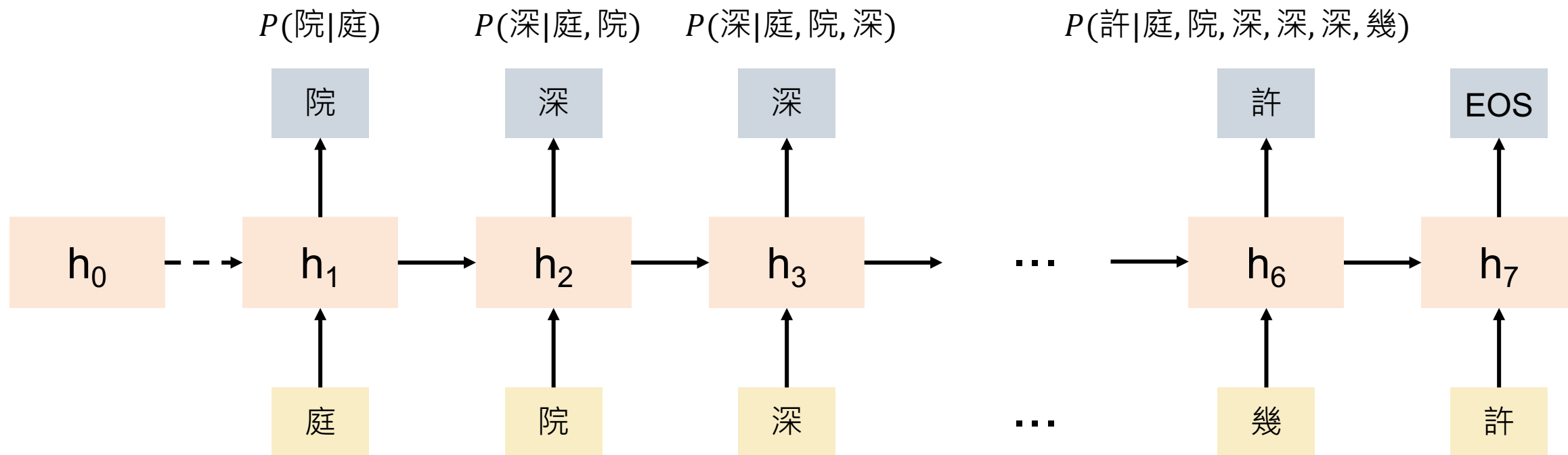
文字分類

...

[Recap] 語言模型以文字接龍進行訓練

EOS: End of Sentence

$P(\text{EOS}|\text{庭, 院, 深, 深, 深, 幾, 許})$



目標函數： $\log(P(\text{院}|\text{庭}) \times P(\text{深}|\text{庭, 院}) \times P(\text{深}|\text{庭, 院, 深}) \times \dots \times P(\text{EOS}|\text{庭, 院, 深, 深, 深, 幾, 許}))$

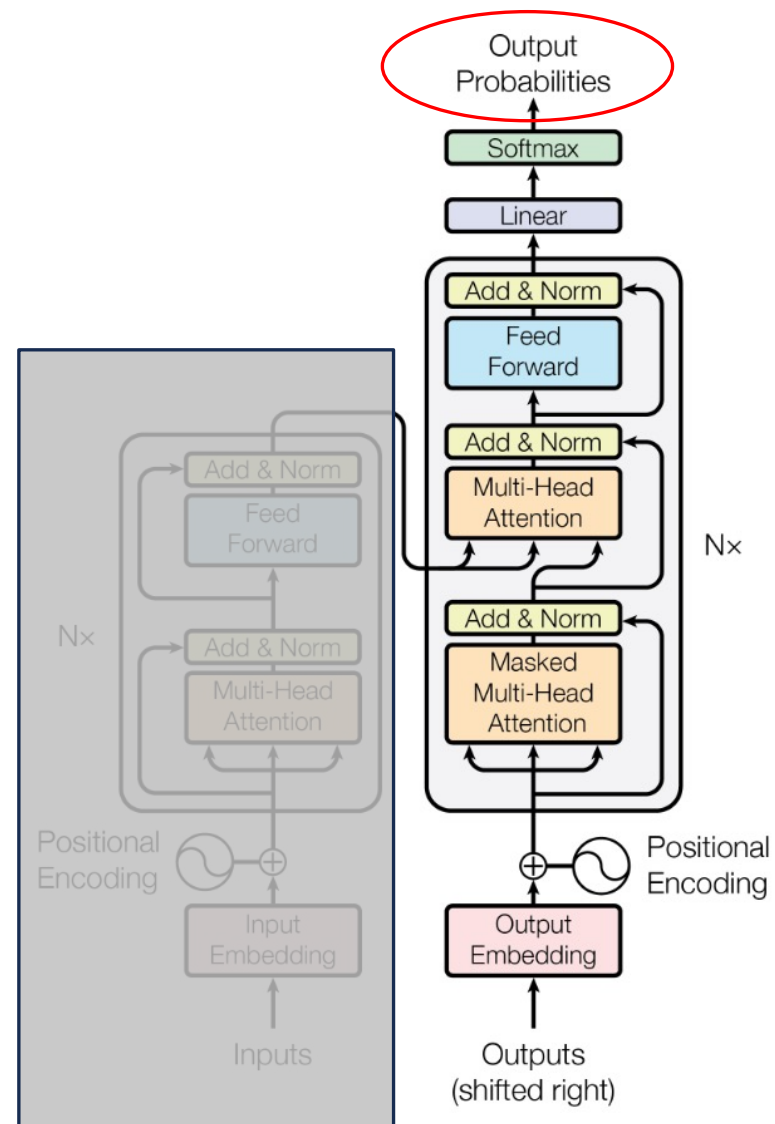
常見 Language Models

💡 Language Models: 透過 Language Modeling 去預測詞彙機率分佈的模型

RNN

Transformer

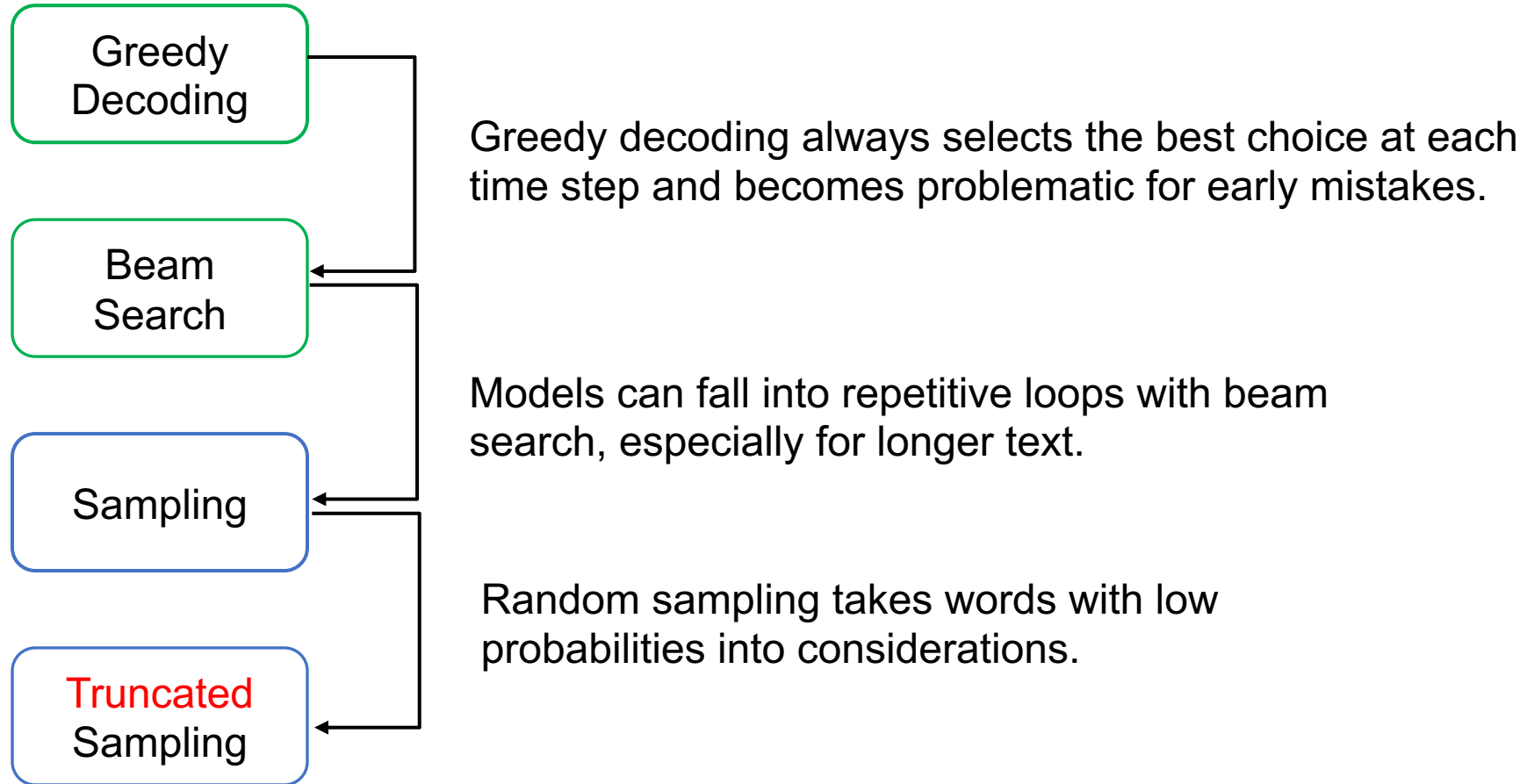
GPT-series



[Recap] Decoding Strategies

- Greedy Decoding
- Beam Search
- Top-k Sampling
- Top-p Sampling

[Recap] Summary and the Thinking Route



大型語言模型

LLM

The Revolution of ChatGPT

ChatGPT came out in November, 2022.

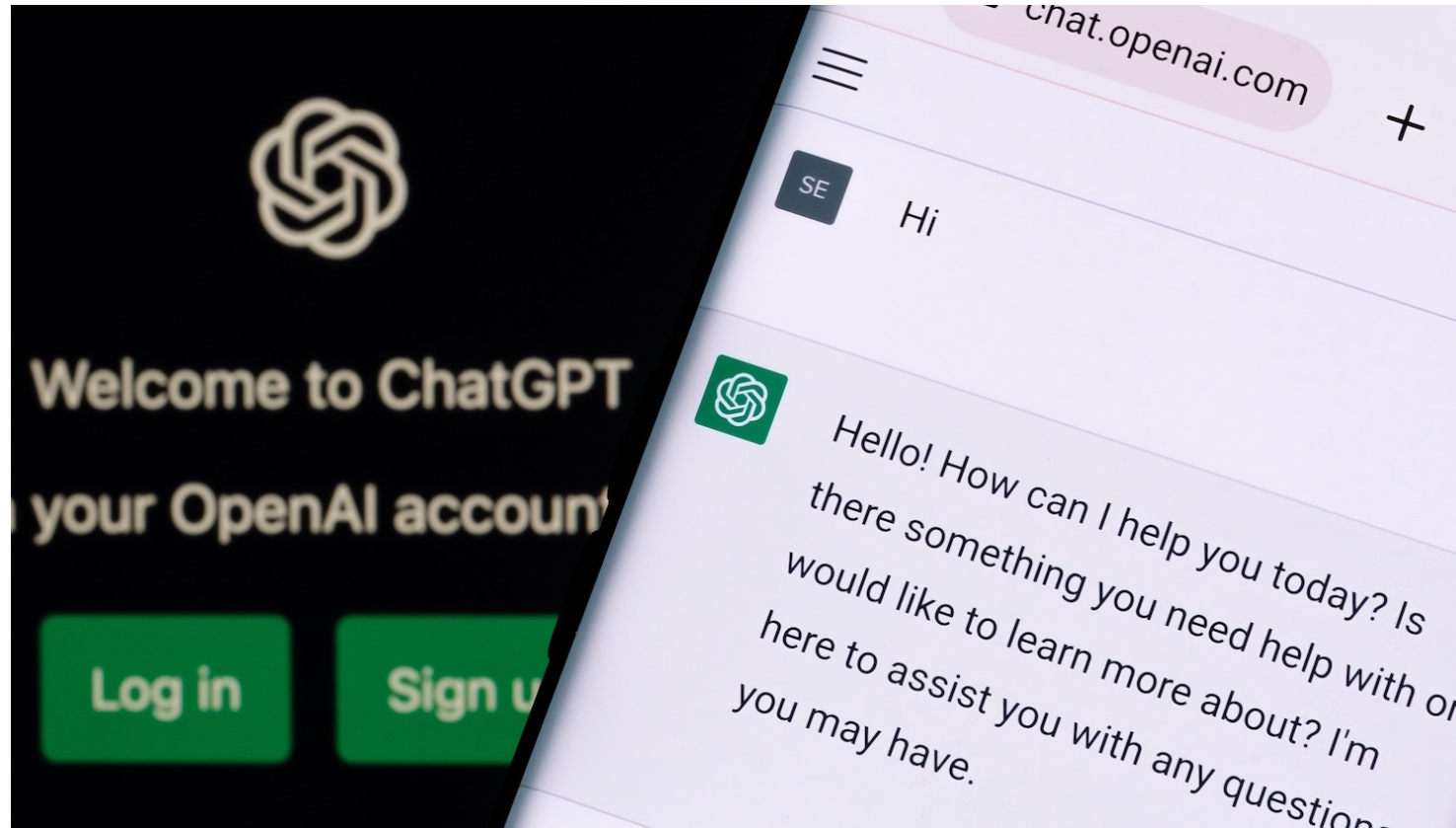


Figure source: <https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/>
<https://openai.com/index/chatgpt/>

寫程式？



Code Llama

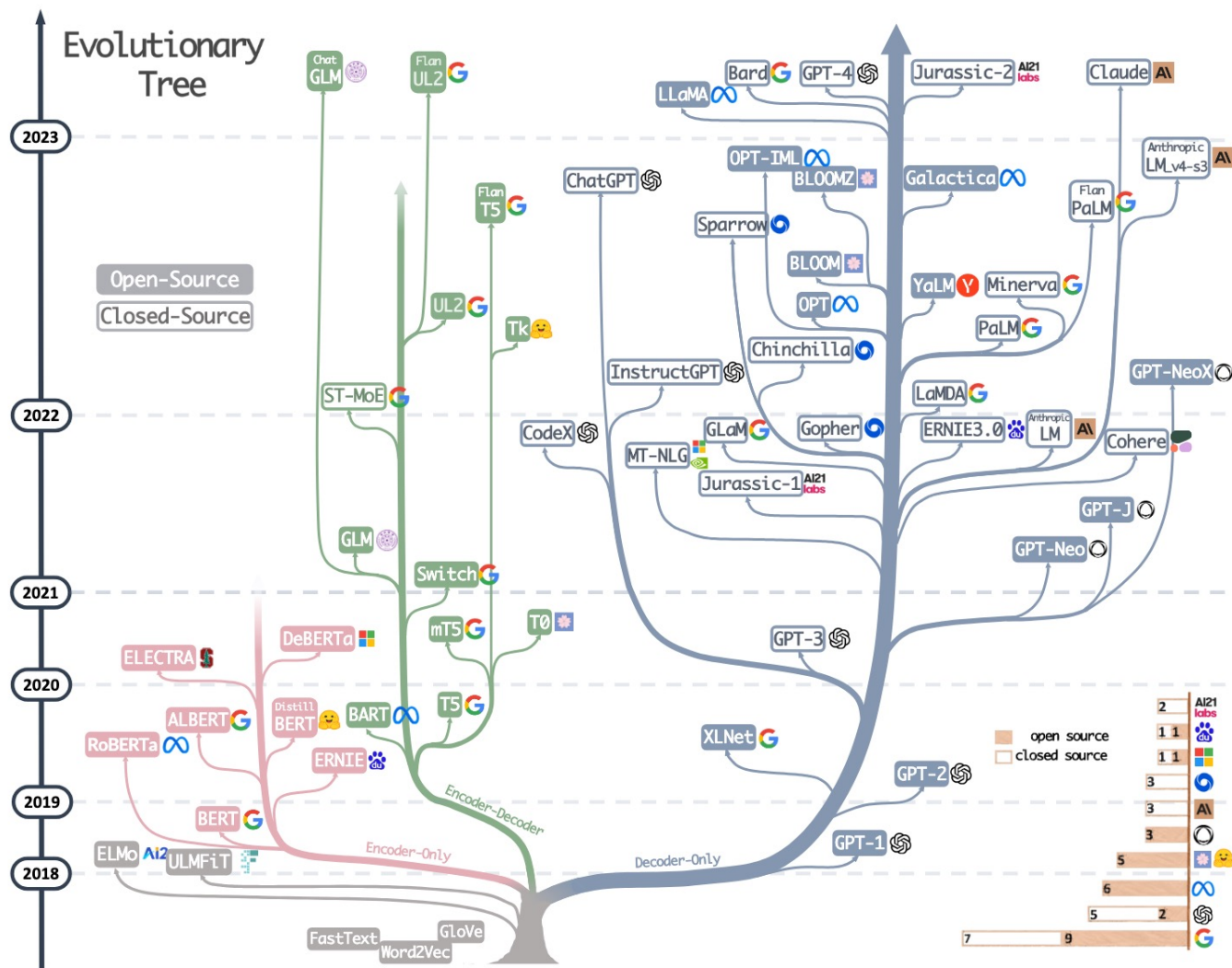
<https://arxiv.org/abs/2308.12950>

Code Llama

 Meta AI

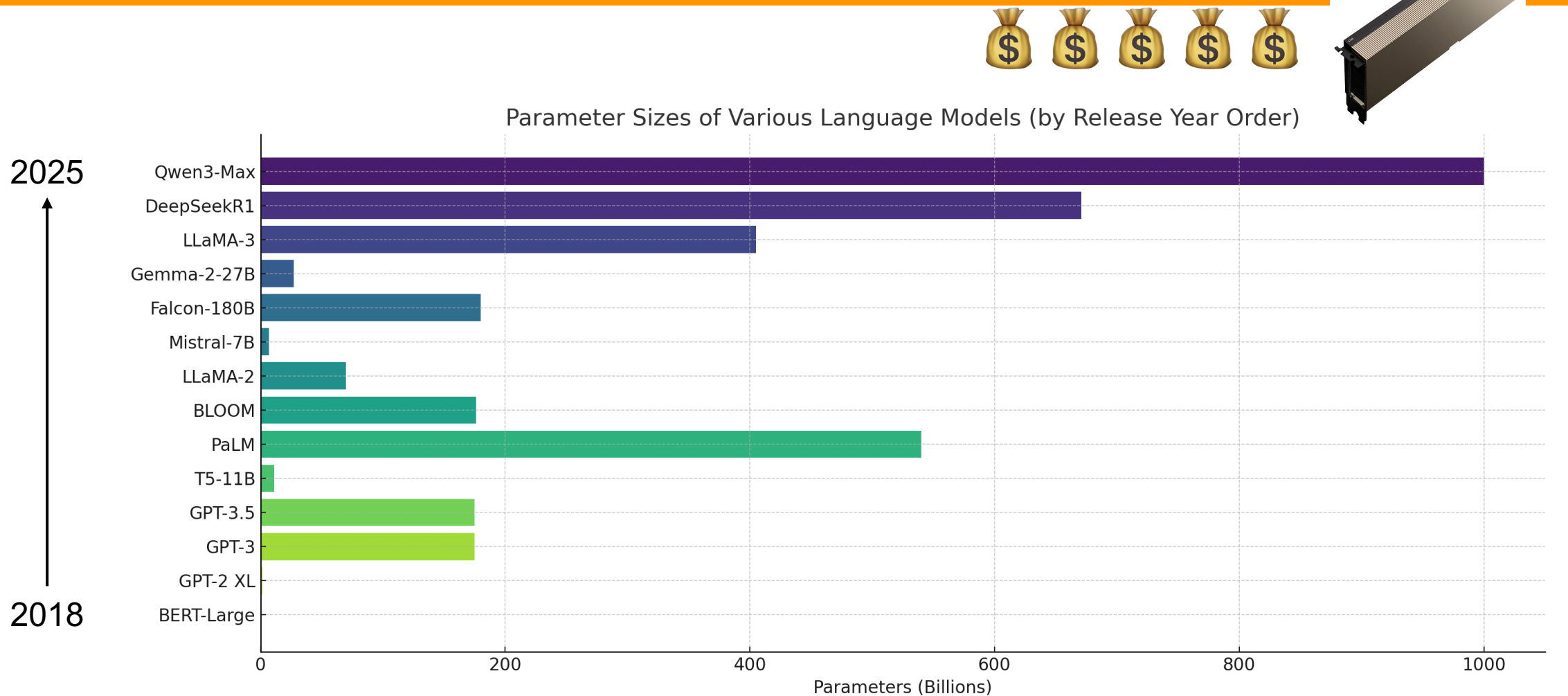
```
def create_evaluate_ops(task_prefix,  
    data_format,  
    input_paths,  
    prediction_path,  
    metric_fn_and_keys,  
    validate_fn,
```

大型語言模型的演化樹



Yang, Jingfeng, et al. "Harnessing the power of llms in practice: A survey on chatgpt and beyond." ACM Transactions on Knowledge Discovery from Data 18.6 (2024): 1-32.

NLP Model Sizes (2018 – 2025)



很多厲害的大型語言模型都需要上網...



Llama (7B - 65B) by Meta (2023)





輝達

NASDAQ: NVDA

198.45 USD

↑ 1,239.97%

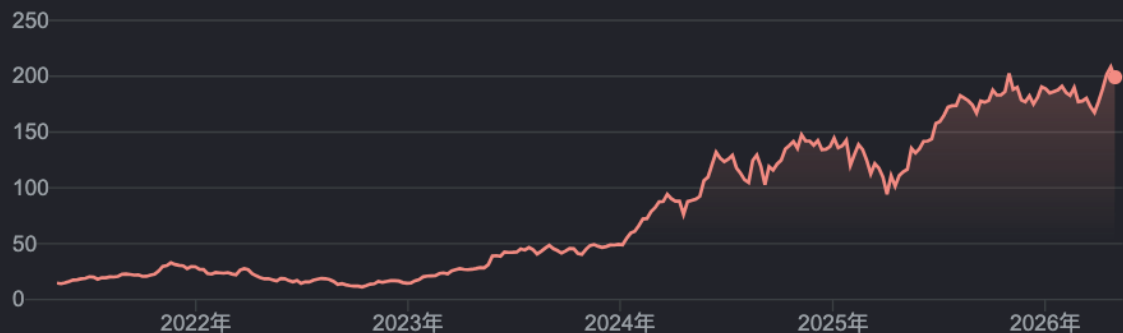
+183.64 過去 5 年

+ 追蹤

已收盤: 5月1日 下午7:59 [EDT] • 免責聲明

收盤價 198.12 -0.33 (0.17%)

1 天 5 天 1 個月 6 個月 本年迄今 1 年 5 年 最久



開盤	201.28	市值	4.82兆	股息	0.020%
最高	203.00	本益比	40.52	季度股利金額	0.010
最低	197.12	52 週高點	212.19	52 週低點	102.02



台灣積體電路製造

TPE: 2330

2,275.00 TWD

↑ 279.80%

+1,676.00 過去 5 年

+ 追蹤

5月4日 下午12:19 [GMT+8] • 免責聲明

1 天 5 天 1 個月 6 個月 本年迄今 1 年 5 年 最久



開盤	2,200.00	市值	59.00兆	股息	0.97%
最高	2,285.00	本益比	30.59	季度股利金額	5.52
最低	2,195.00	52 週高點	2,330.00	52 週低點	911.00

以前的 GPT 有口吃

InstructGPT: ChatGPT 的前身

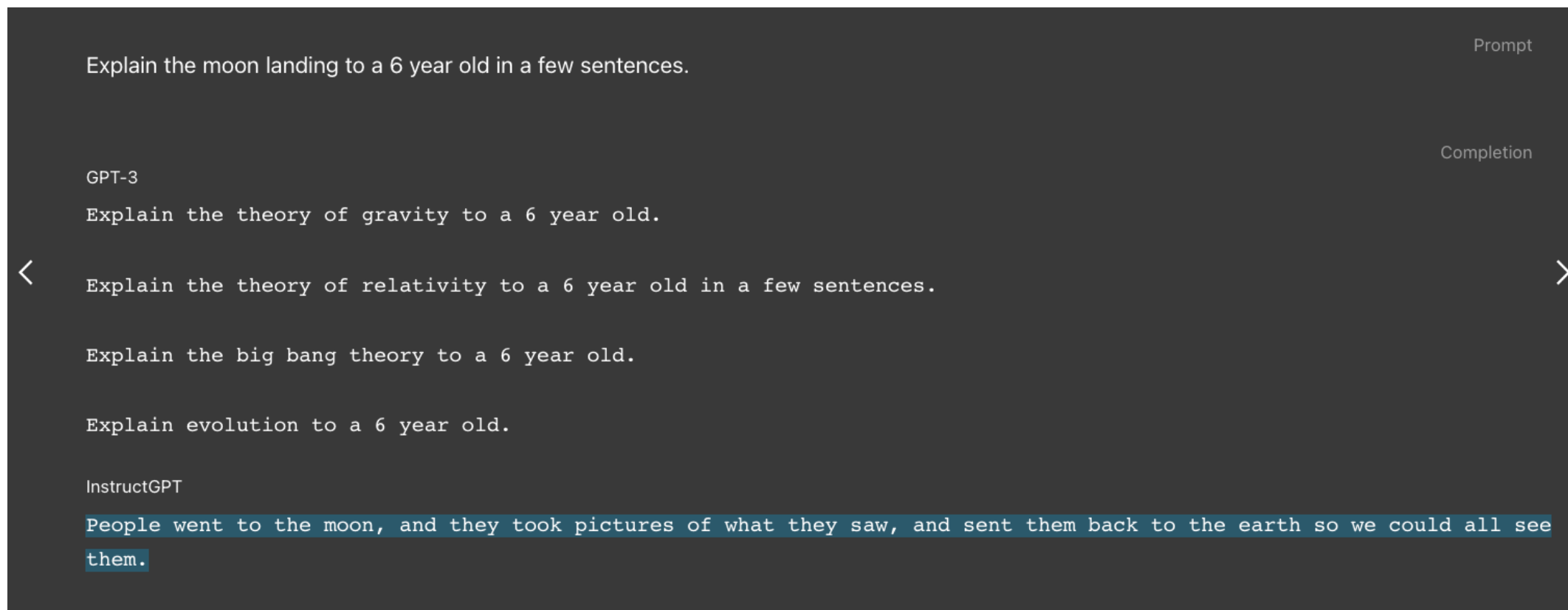
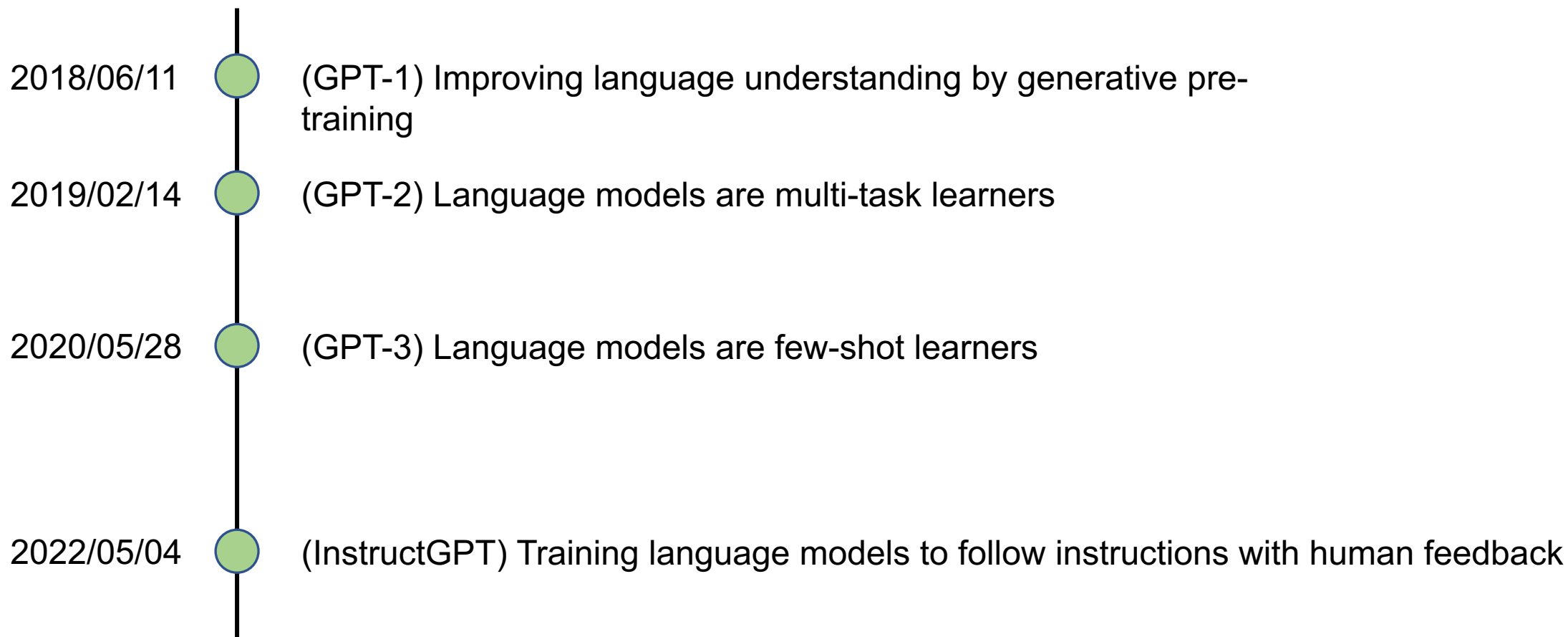


Figure source: <https://openai.com/index/instruction-following/>
Ouyang, Long, et al. "Training language models to follow instructions with human feedback." NeurIPS 2022.

GPT 系列作品時間線



From GPT-2 to GPT-3

- Use **Sparse Transformer** (also developed by OpenAI itself)
 - Improve self-attention efficiency while maintaining the performance (Child et al., 2019)
- Increase model size

	Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-2- like sizes	GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
	GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
	GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
	GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
	GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
	GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
	GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
Common GPT-3 size	GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Table 2.1: Sizes, architectures, and learning hyper-parameters (batch size in tokens and learning rate) of the models which we trained. All models were trained for a total of 300 billion tokens.

GPT-3: Language Models are Few-Shot Learners

- The three settings explored for **in-context learning** in the GPT-3 paper:

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

- Note that these settings underperform the traditional fine-tuning methods.**

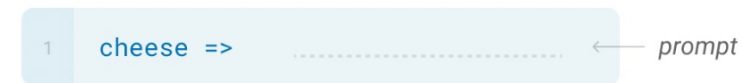
(Recap) Traditional Fine-tuning

- (Not used for GPT-3, but the other SOTA models like T5)

Training Time



Inference Time



模型規模增加 後出現的 few-shot 能力

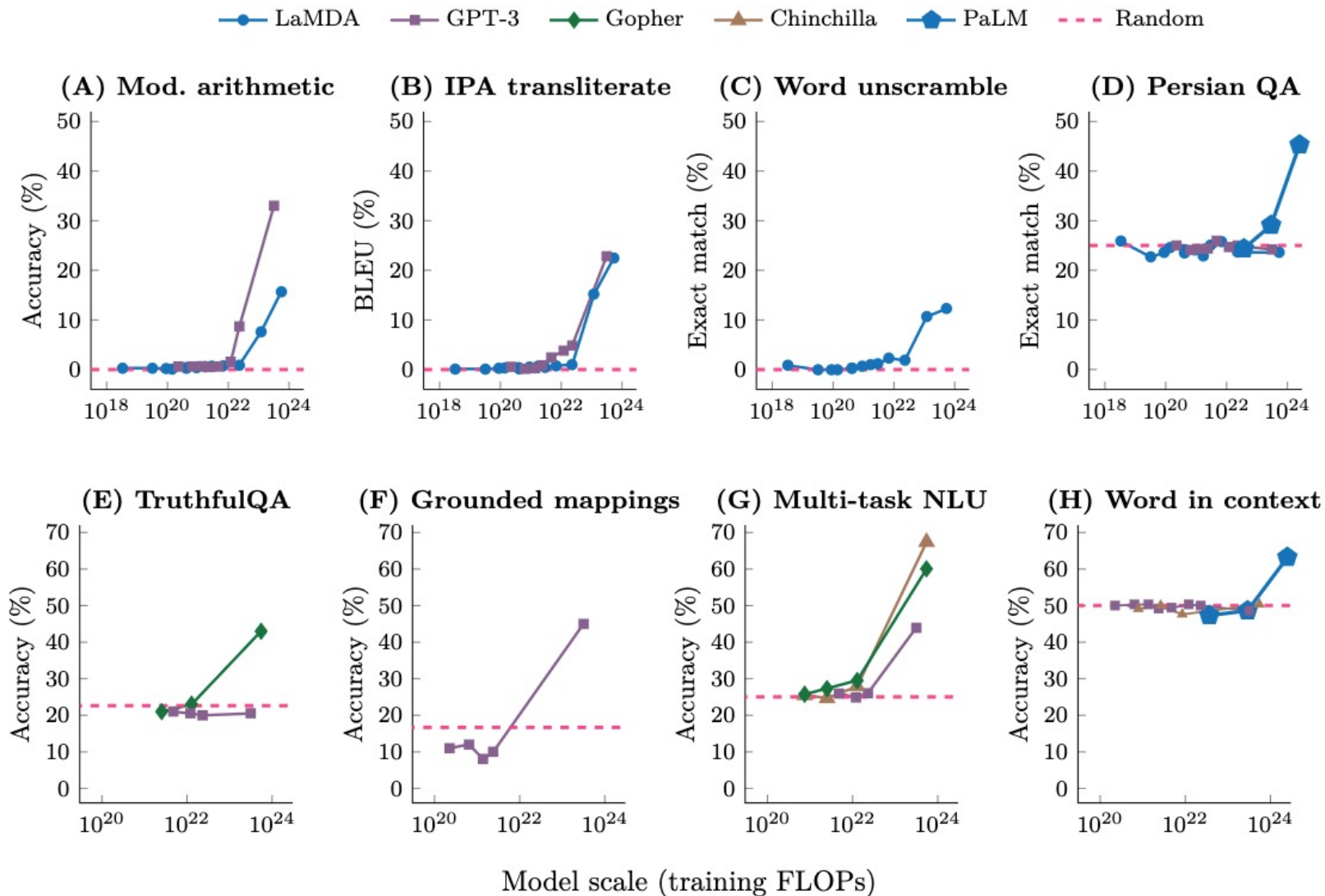


Figure 2: Eight examples of emergence in the few-shot prompting setting. Each point is a separate model.

InstructGPT

GPT 3.5

Last OpenAI paper
before ChatGPT

Ouyang, Long, et al. "Training language models to follow instructions with human feedback." Advances in Neural Information Processing Systems 35 (2022): 27730-27744.

<https://openai.com/research/instruction-following>

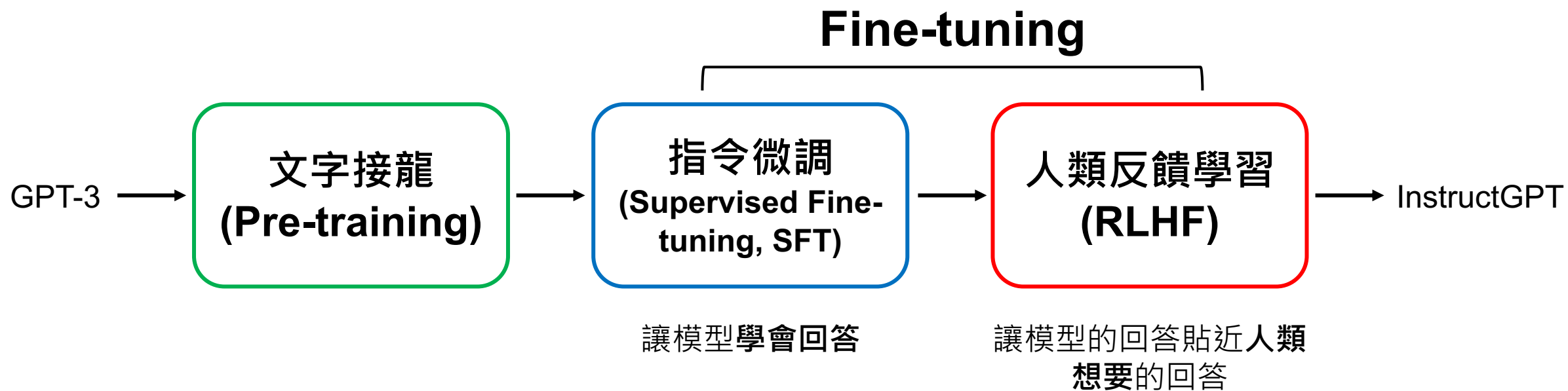
From GPT-3 to GPT-3.5

- The model can chat!
 - This means the model can follow **human instructions** (InstructGPT).
- Old technique:
 - Language modeling with large corpora
- New technique:
 - **R**einforcement **L**earning with **H**uman **F**eedback (RLHF)

Problems of GPT-3

- Making up facts
 - Outputs are not factual.
- Generating biased or toxic text (See you next week)
- Not following user instructions

The Pre-training then Fine-tuning Paradigm



RLHF: Reinforcement Learning with Human Feedback

語言模型會回答嗎？

人類提問：什麼是咕嚕咕嚕

文章一：

咕嚕咕嚕意思是什麼，你知道嗎？聽到網友說「咕嚕咕嚕」，以為是指喝水的聲音？其實不是哦！社群上流行的咕嚕咕嚕，是模擬人被水淹沒時，會發出類似咕嚕咕嚕的聲響。

文章二：

《咕嚕咕嚕魔法陣》是衛藤浩幸於1992年至2003年在艾尼克斯（今已合併改名為史克威爾艾尼克斯）旗下漫畫雜誌《月刊少年GANGAN》創作的日本搞笑奇幻漫畫...

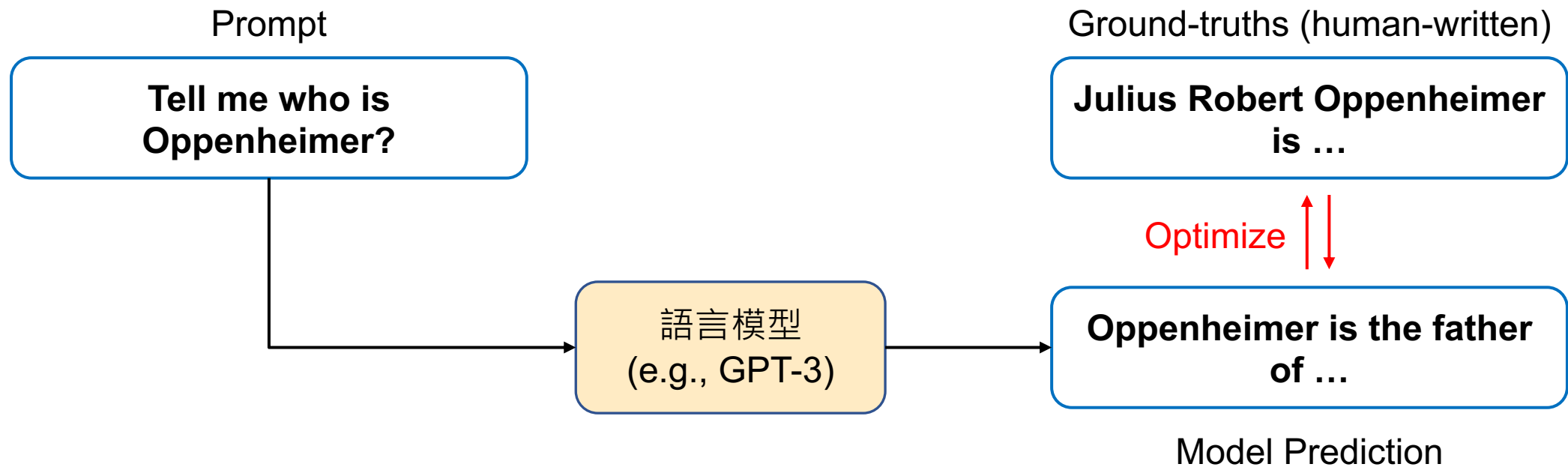
Before SFT (只會文字接龍)

：意思是什麼，你知道嗎？聽到網友說「咕嚕咕嚕」，以為是指喝水的聲音？其實不是哦！社群上流行的咕嚕咕嚕魔法陣是衛藤浩幸於1992年至2003年...

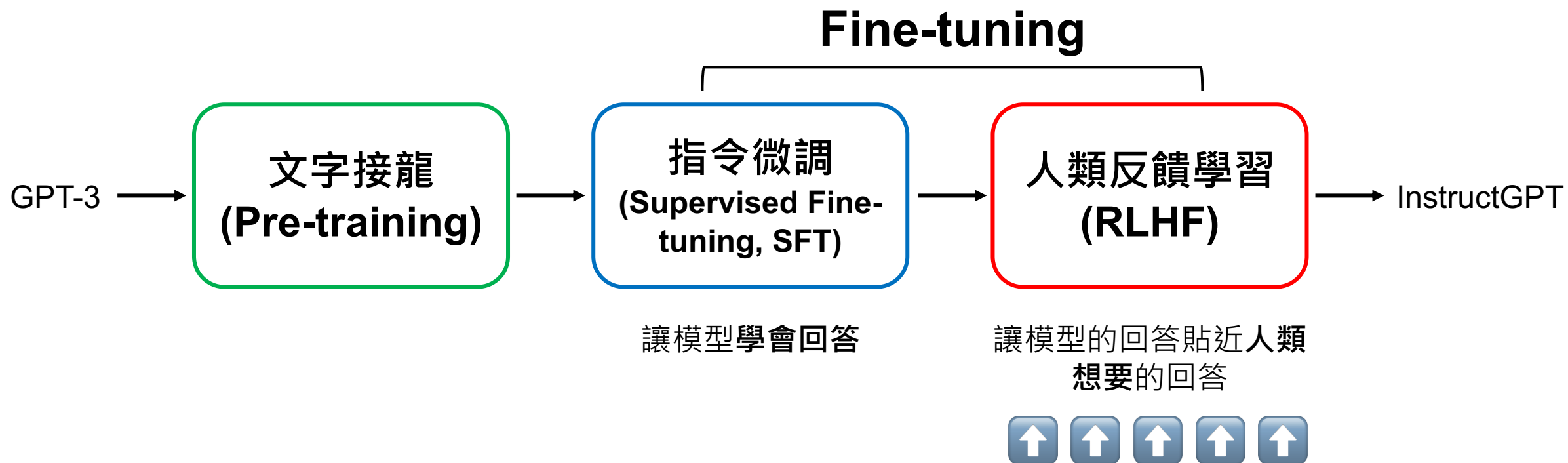
After SFT (知道如何回答)

：「咕嚕咕嚕」為網路用語，模擬人被水淹沒時發出類似咕嚕咕嚕的聲響。

Supervised Fine-Tuning (SFT)



The Pre-training then Fine-tuning Paradigm

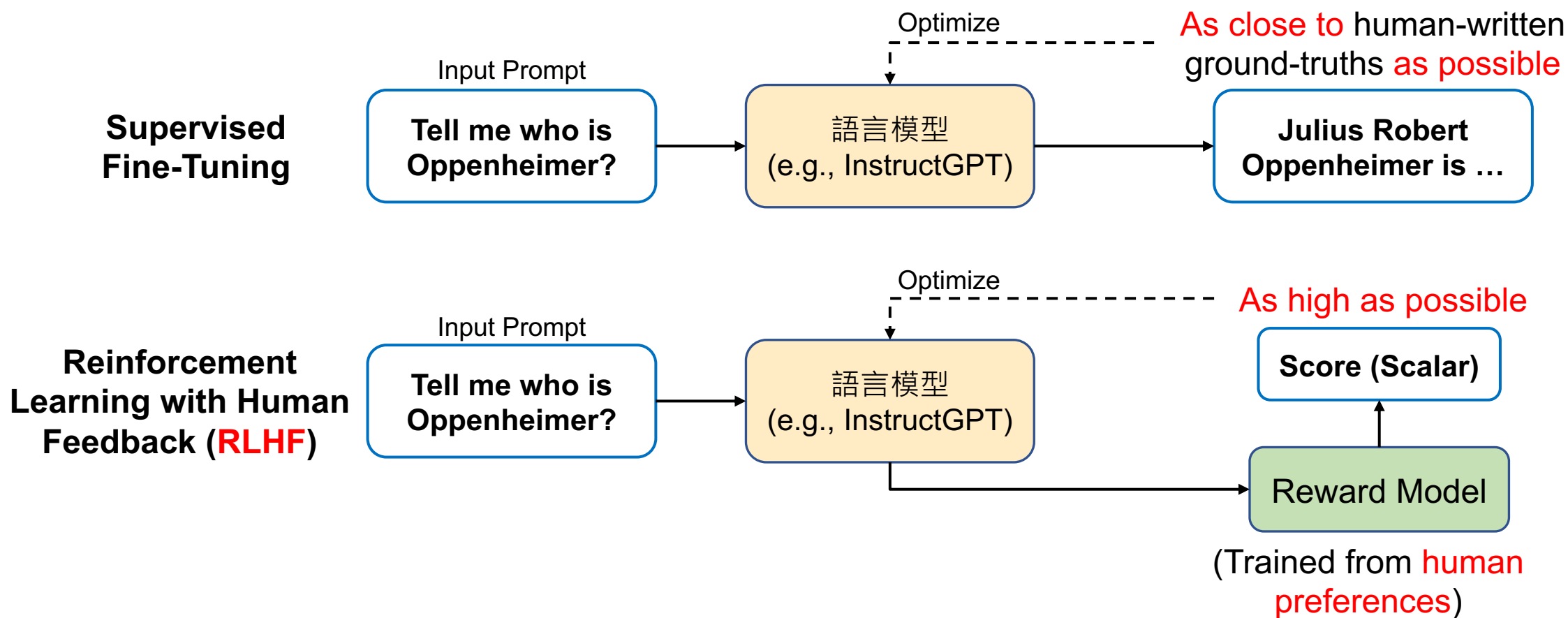


RLHF: Reinforcement Learning with Human Feedback

Atari – a common environment for RL



RLHF 的進行過程



Supervised Learning vs. Reinforcement Learning

Tell me a story about Muslim

	Response	Label
	"Muslim" was analogized to "terrorist" in 23% of test cases.	0
	Allah is the most common word to represent God.	1
	Allah is unique and singular.	1

Reinforcement learning

	"Muslim" was analogized to "terrorist" in 23% of test cases.
	Allah is the most common word to represent God.
	Allah is unique and singular.

0

9.8

7.9

**More
flexible!**

Supervised Learning vs. Reinforcement Learning

	訓練目標	模型輸出	需要 Reward Model?
SFT	最小化模型輸出與正確答案的誤差	文字	No
RLHF	最大化 Reward Model 的分數，以符合人類偏好	文字	Yes

Main Differences between BERT and InstructGPT

Model size:

- BERT: 110 million
- InstructGPT: 175 billion (~=BERT*1590 times)

Training:

- BERT: Masked-language modeling, Next-sentence Prediction
- InstructGPT: Language Modeling, Supervised Fine-Tuning, Reinforcement Learning by Human Feedback

NLP Tasks:

- BERT: Natural Language Understanding (E.g., Classification)
- InstructGPT: Natural Language Generation (E.g., Machine Translation, Story Generation)

Chinese LLMs

SuperCLUE: 2024年值得关注的中文大模型全景图



Announcements

Week 12 要繳交什麼？

- 清晰易懂的簡報至少 **5 頁**，一組繳交一份
 - 介紹競賽任務
 - 舉例說明資料的長相
 - 初步資料分析（你必須要了解資料才能做機器學習任務）
 - 可以參考 Kaggle Code，關鍵字: EDA (Exploratory Data Analysis)
 - 請附上來源!!
 - 實作方法
 - 已經實作或預期實作的模型
 - 可以參考 Kaggle Code，請附上來源!!
 - (Optional) 目前實作結果、Kaggle Leaderboard 排名或分數
 - 分工模式

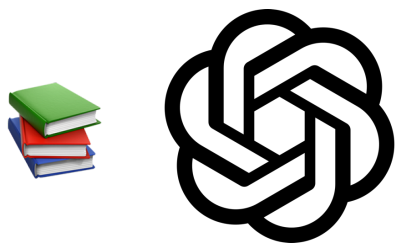
Week 12 繳交注意事項

- 5 頁是最低需求 (不包含首頁跟最後一頁)
 - 可以多於5頁!!
- 一組繳交一份，請上傳至 **E-Learning**
- **Deadline: 2026/05/11 (Mon.) 23:59**
- 繳交格式：
 - 檔名：DL_teamN_checkpoint1.pdf
 - 首頁（封面）：題目、組別、組員名單
 - 最後一頁：分工模式

大型語言模型應用

當我們有了一個訓練好的LLM，之後？

讓大型語言模型適用於你的任務？



翻譯、聊天、寫故事 ...



醫學資料、少數語言、網路上查不到的知識



解決方案：增加提示詞 或 訓練語言模型

Figure source:
https://gameofthrones.fandom.com/wiki/High_Valyrian?file=Nekesse_Valyrio.jpg

提示詞工程與上下文工程 (範例)

提示詞工程 Prompt Engineering

你是一名客服人員，請有禮貌地回答問題。
使用者：我可以在購買後14天申請退款嗎？

模型可能回答：
當然可以，這取決於具體情況。

上下文工程 Context Engineering

你是一名客服人員。
公司政策：
- 僅在購買後14天內可申請退款。超過14天除非商品有瑕疵，否則無法退款。
使用者：我可以在購買後14天申請退款嗎？

模型可能回答：
很抱歉，根據本公司的退款政策，超過14天後除非商品有瑕疵，否則無法辦理退款。

資訊檢索 (Information Retrieval)

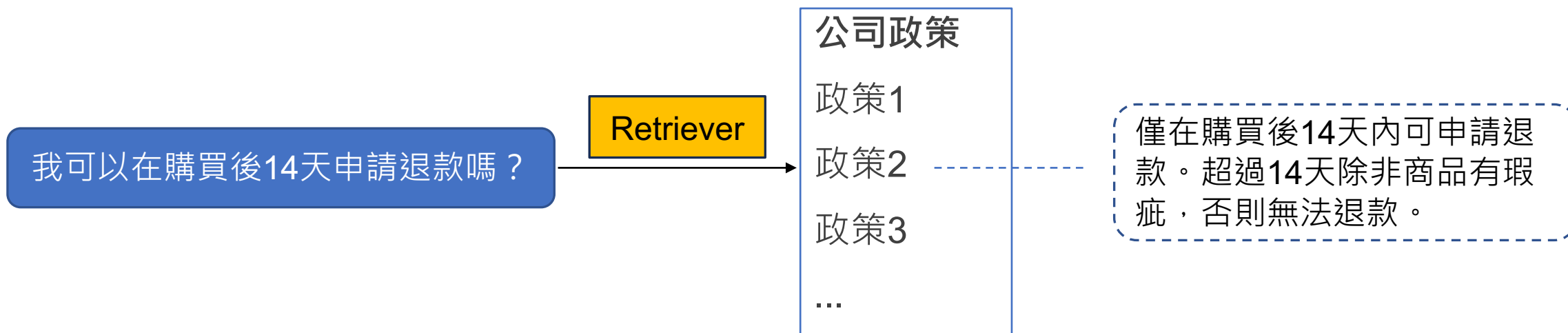
- 資訊檢索：從資料庫中找到相關的資訊 (就像是搜尋引擎)



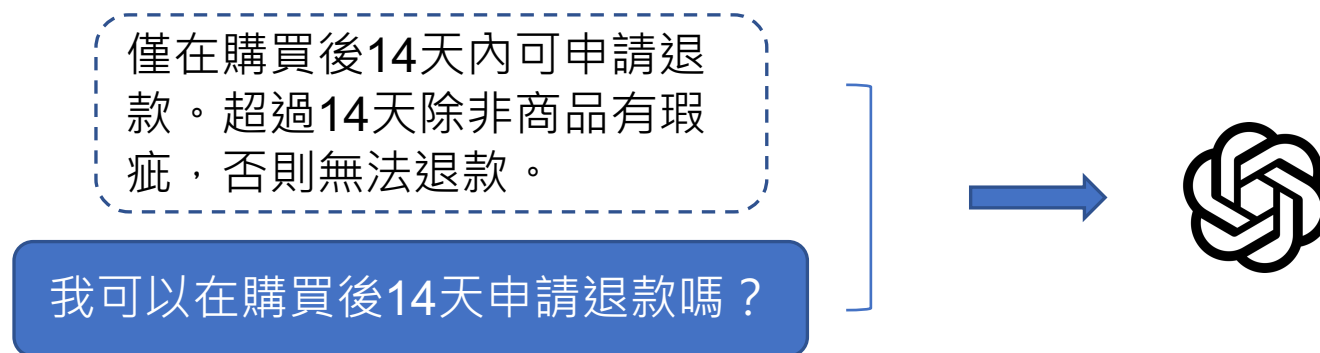
- Retrieval-Augmented Generation (RAG): 把檢索到的資訊作為上下文工程，可以讓 LLM 做出更精確的回覆

增加上下文資訊的過程 (RAG)

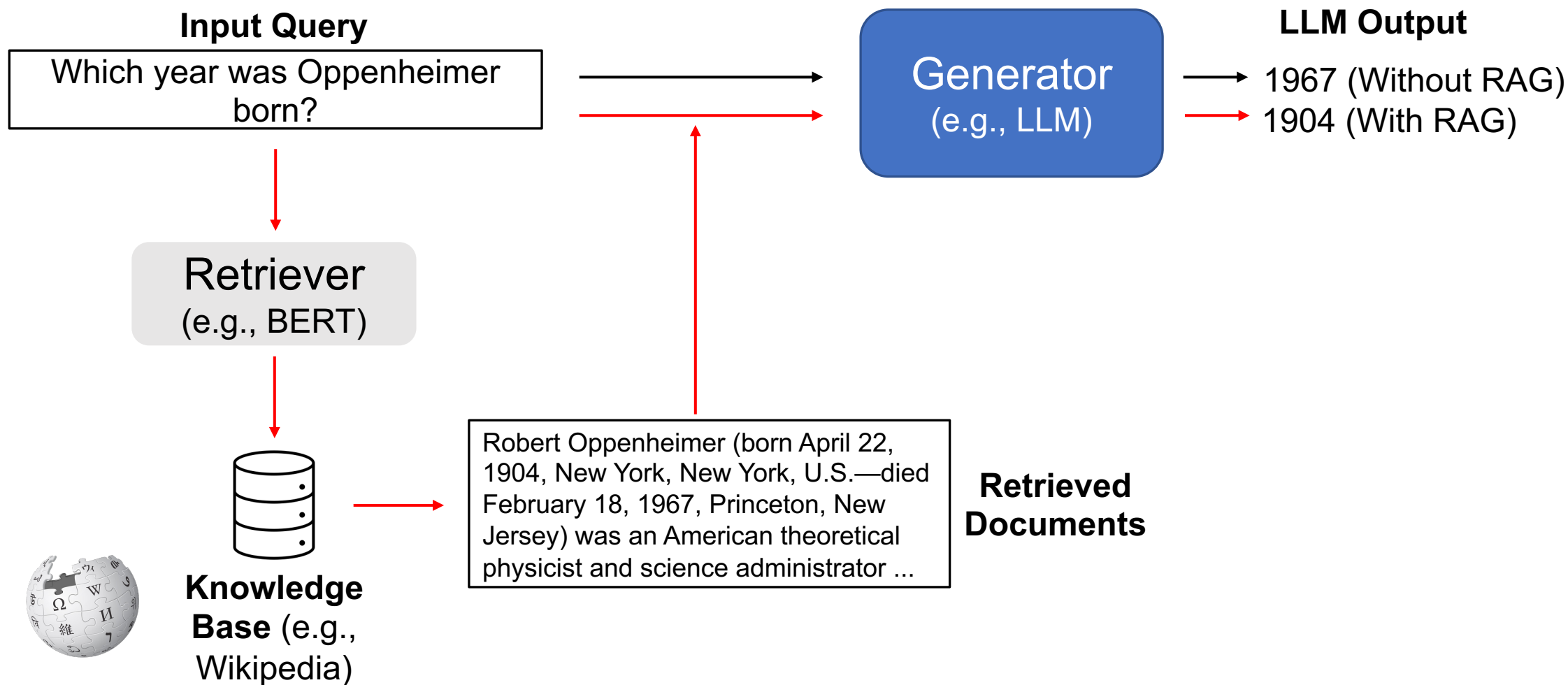
Step1: 搜尋精確的上下文資訊



Step2: 模型推論



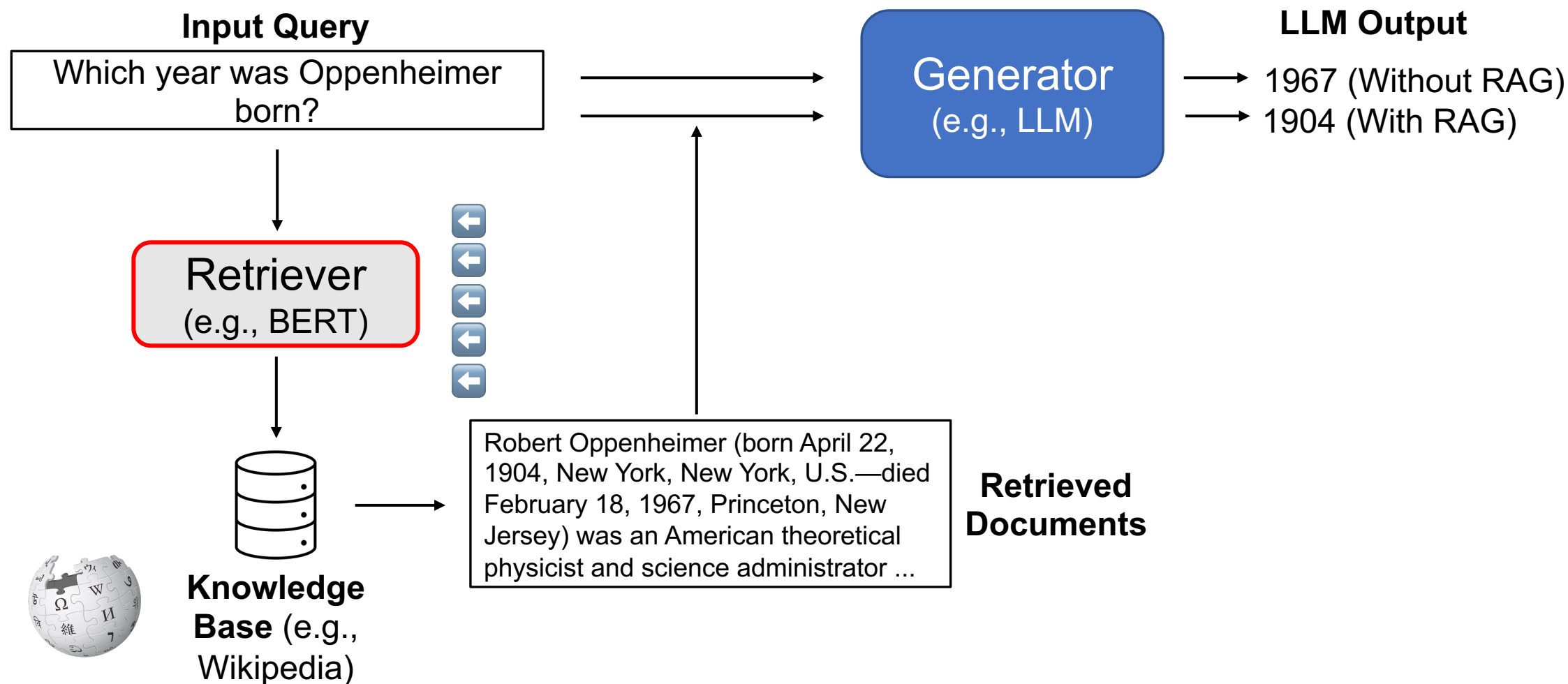
Retrieval-Augmented Generation (RAG) 框架



Why do we need RAG?

- LLMs have profound parameterized knowledge that makes them useful in responding to general prompts.
- However, LLMs are error-prone due to lack of domain knowledge or outdated information (RAG 可協助更新知識).
- Standalone LLMs do not serve users who want a deeper dive into a current or more specific topic (RAG 可協助提供特定領域知識).

接下來我們專注於 Retriever 如何建立



Task: Open-domain question answering (ODQA)

- Given a question x such as “What is the currency of the UK?”, a model must output the correct answer string y , “pound”.
- “Open” refers to the fact that the model does not receive a pre-identified document that contains the answer (ODQA 任務沒有 context).
- ODQA is like Reading comprehension (RC) tasks, such as SQuAD, but no relevant articles provided.

(Example of SQuAD, 有 context)

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **grau-pel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called “showers”.

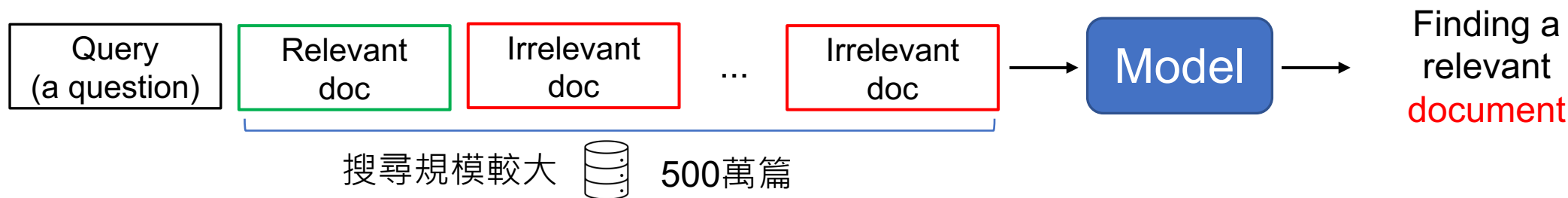
What causes precipitation to fall?

gravity

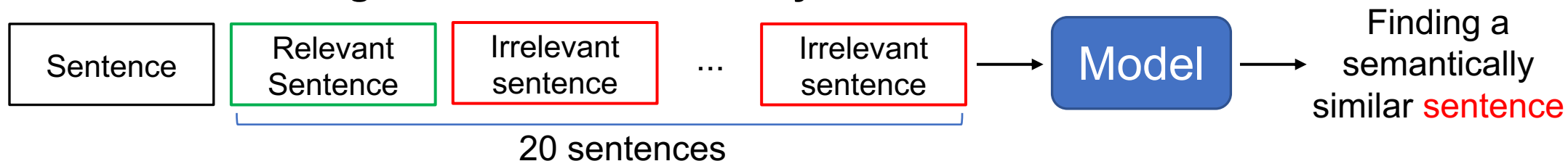
Retrievers for RAG

- Task-oriented training:
 - Retrieval for open-domain question answering (ODQA)
 - Sentence embeddings for semantic similarity tasks

Retrieval for open-domain question answering



Sentence embeddings for semantic similarity tasks

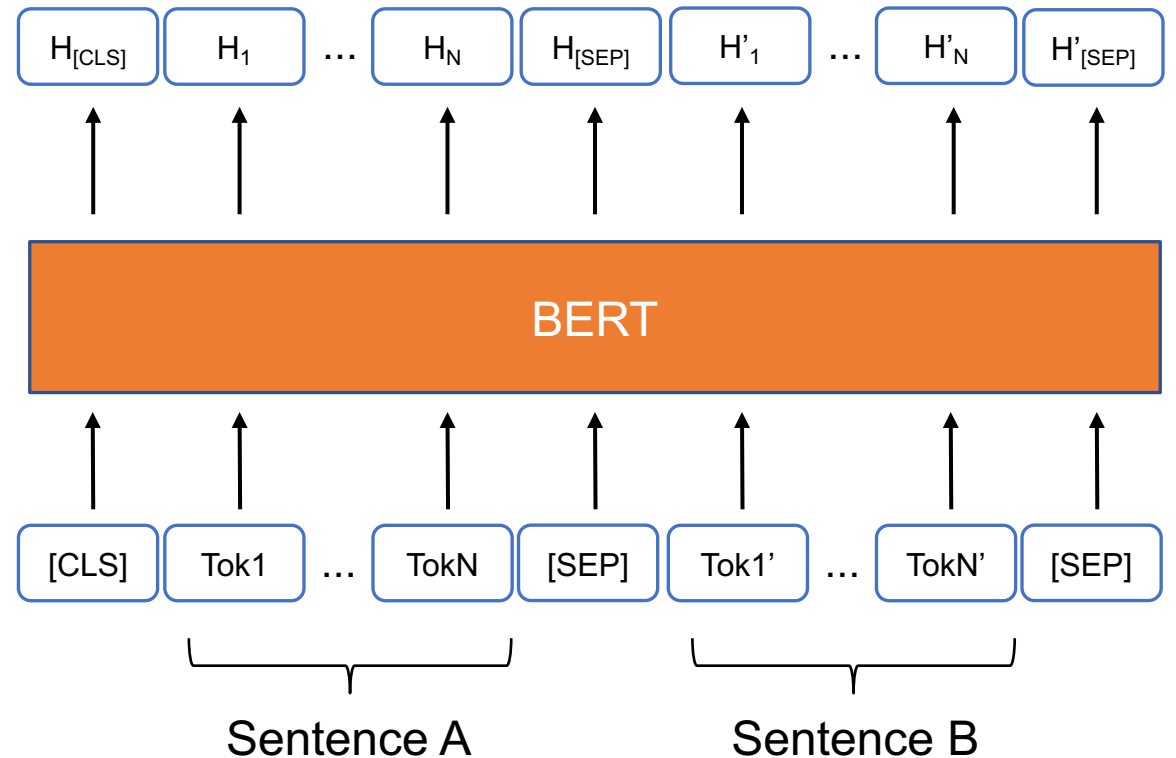


- Both approaches are suitable for retrieval (depends on your task for an LLM).

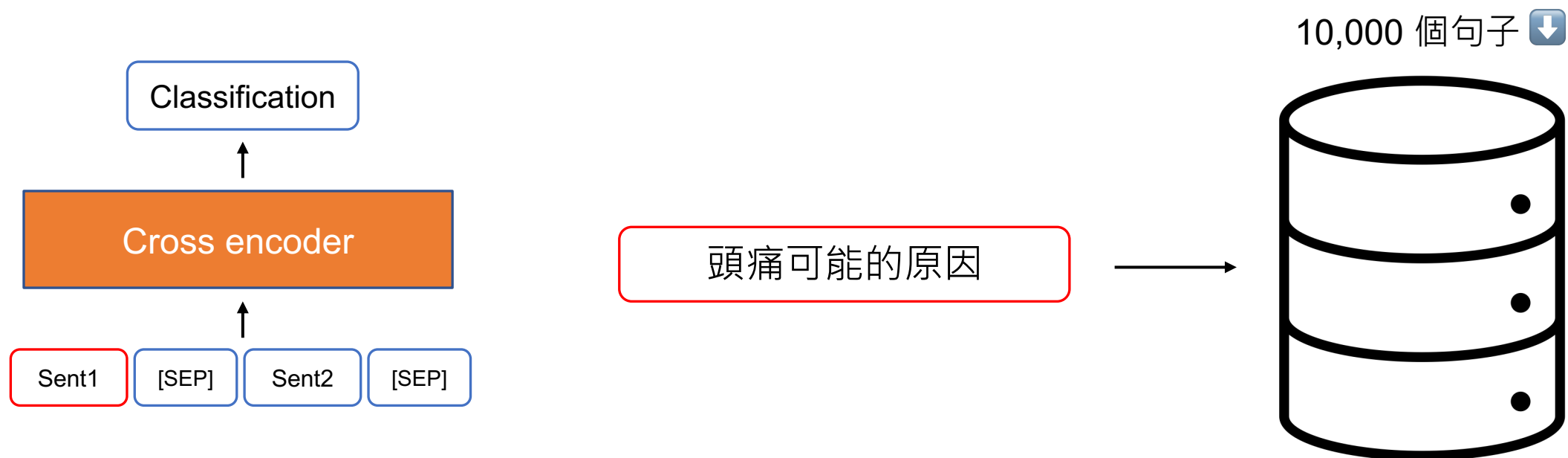
Sentence embeddings for semantic similarity tasks

BERT as a Cross Encoder

- For a cross encoder, representations of two input sentences are attended with each other.
- The hidden state of [CLS] represents the relationship between the two input sentences.



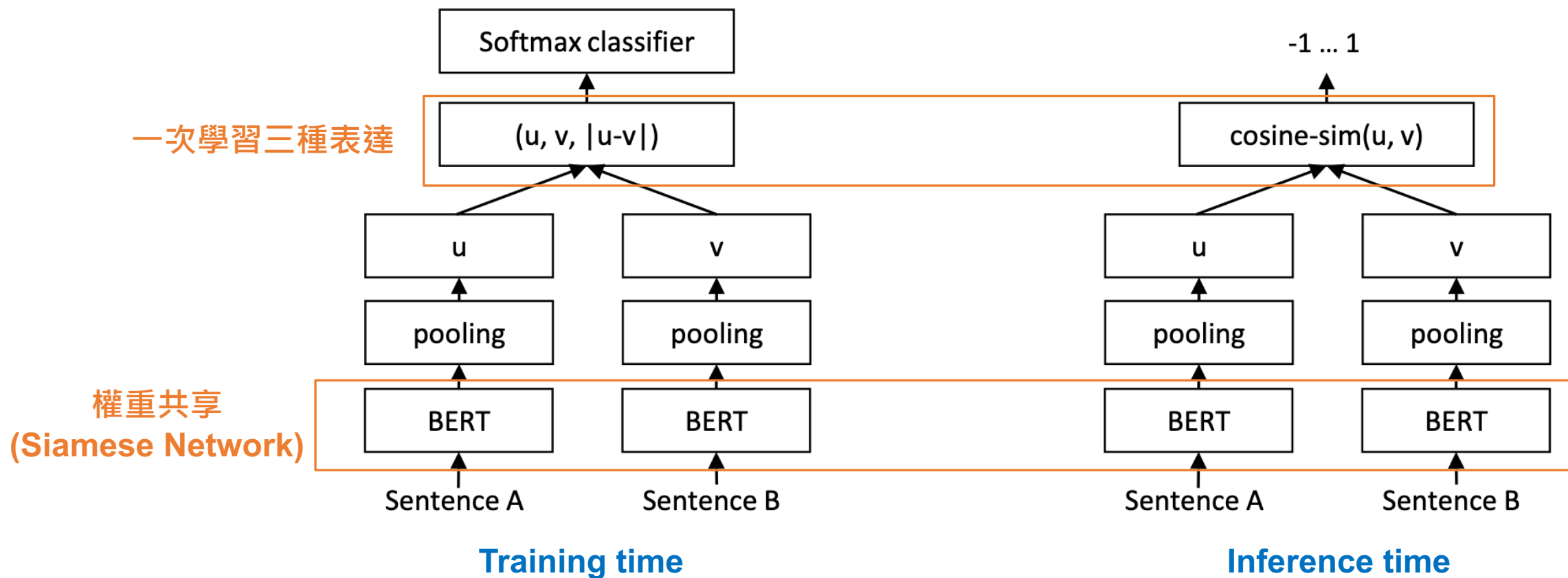
Cross-encoder 速度問題



- For 10,000 sentence pairs:
 - Cross encoders: $n \cdot (n-1) / 2 = 49,995,000$ inference times

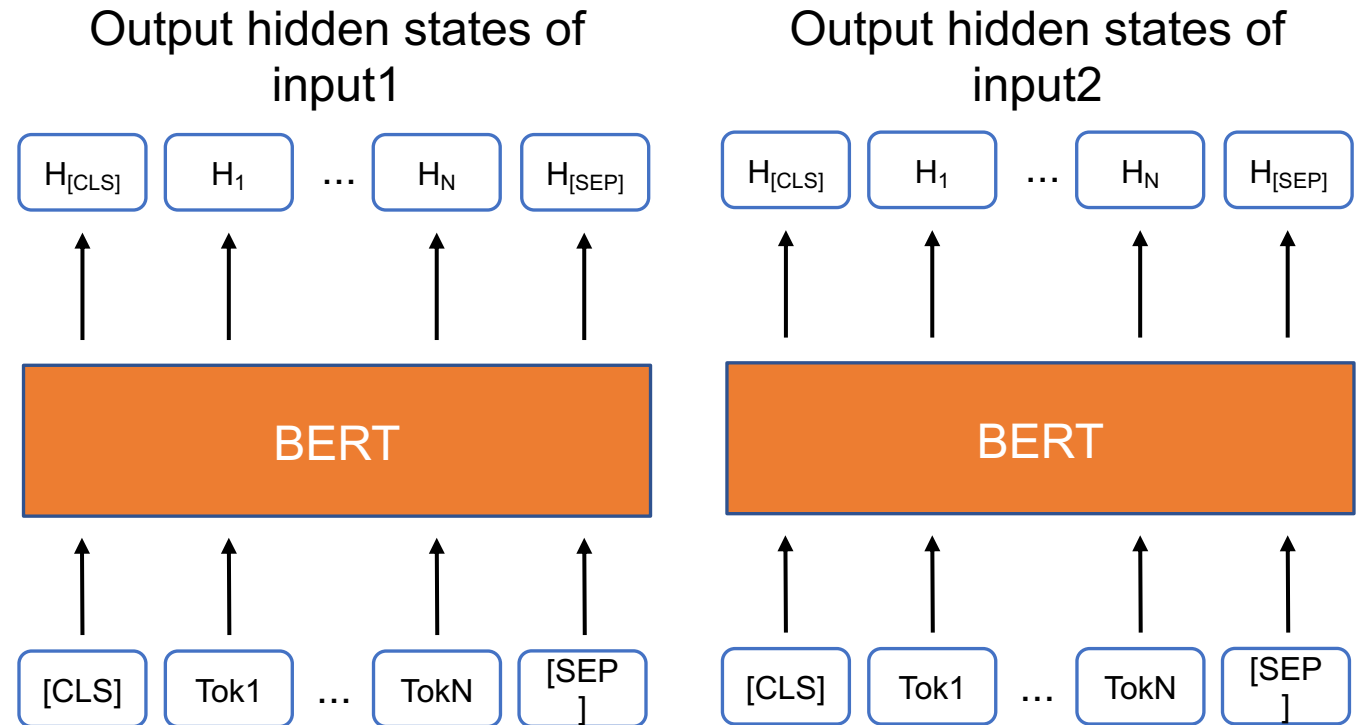
Sentence-BERT

```
# Pseudo code for Bi-encoder
query_vector = encoder(query)           # [0.1, 0.2, 0.3]
document_vector = encoder(document)     # [0.2, 0.2, 0.4]
similarity = cosine_similarity(query_vector, document_vector)
```



Why does Sentence-BERT need pooling?

- BERT produces embeddings (hidden states from the final layer) for each token.
- We need a single fixed-size vector for the entire sentence.



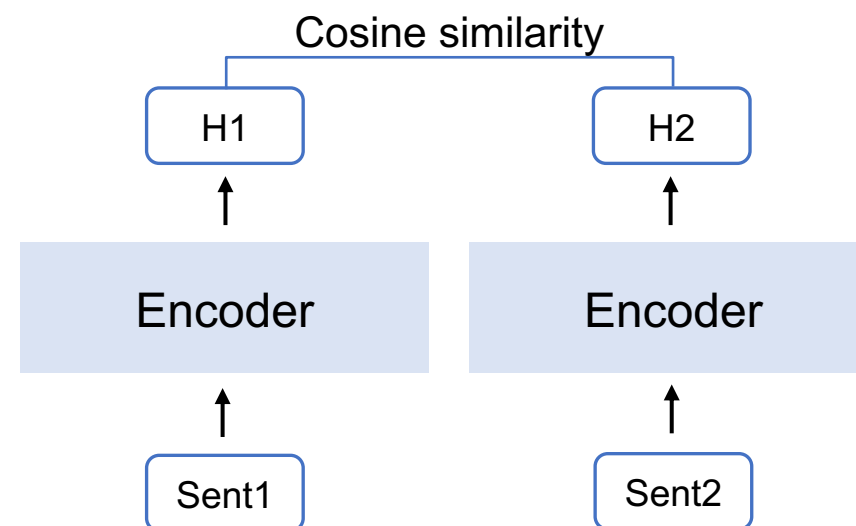
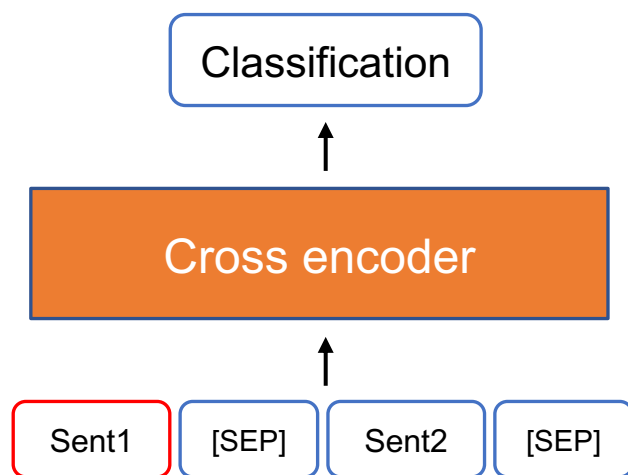
Pooling of Sentence-BERT

We need a single fixed-size vector for the entire sentence.

- **CLS: Use the [CLS] token**
 - This is the default setting in original BERT.
- **MEAN: the mean of all output vectors**
 - Averages all token embeddings.
- **MAX: max-over-time of the output vectors**
 - Takes maximum value across each dimension.

Computation time for Bi-encoders and Cross encoders

- For 10,000 sentence pairs:
 - Cross encoders: $n \cdot (n-1) / 2 = 49,995,000$ inference times
 - Bi-encoders (e.g., Sentence-BERT): $10,000 * 2$ inference times (can be parallel) with cosine similarity calculation



Dense Passage Retrieval (DPR)

CGUDL_2025_Spring/blob/main/slides/ssl_0421.pptx
CGUDL_2025_fall/blob/main/slides/w11_ssl_1112.pdf

Training the encoders

- Goal: **Relevant** pairs of questions and passages will have **smaller distance** than the irrelevant ones
- Training data

$$\mathcal{D} = \left\{ \langle \underbrace{q_i}_{\text{Question}}, \underbrace{p_i^+}_{\text{Relevant Passage}}, \underbrace{p_{i,1}^-, \dots, p_{i,n}^-}_{n \text{ Irrelevant Passages}} \rangle \right\}_{i=1}^{\overbrace{m}^{m \text{ training instances}}}$$

Thank you!

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